

 1. INTRODUCTION TEAM

Student Attendance System

Object-Oriented Python Program for Classroom Attendance & Analytics

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Problem Analysis & Project Lead



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OOP Architecture & Methods



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Testing & Data Integration

Introduction: Object Problem & Analysis



🕒 Time Loss on Roll Calls

Manual paper attendance wastes 10–15 minutes of lesson time daily.

📊 Manual Math Errors

Calculating percentage attendance rates manually causes frequent mistakes.

📄 Data Vulnerability

Physical rosters can be lost, damaged, or modified without authorization.

Content Object: Classroom Case Example

Scenario: A teacher manages a class of 30 students across multiple sessions. In a manual system, calculating each student's attendance percentage requires manual counting and division.



1. Create Student Object

Assign ID STU-101 to student "Alex Smith".



2. Record 3 Sessions

Logged Present for Class 1 & 2, Absent for Class 3.



3. Compute Rate

Encapsulated formula outputs **66.67%** instantly.

Solution: Automated Student Attendance System

👤+ Add Student Information

Instantiates student profiles with ID and Name using OOP constructors.

📝 Record Attendance

Updates total classes and present counts via method parameters.

📊 Calculate Attendance Rate

Automates percentage formula with zero mathematical errors.

🛡️ Encapsulation Safeguard

Private attributes (`__student_id`, `__attendance`) prevent external tampering.

💻 Simple User Interface

Clean interactive console prompts for fast classroom entry.

Code Preview: Python Class & Execution

```
class Student:
    def __init__(self, student_id, name):
        self.__student_id = student_id
        self.__name = name
        self.__attendance = 0
        self.__total_classes = 0

    def record_attendance(self, present):
        self.__total_classes += 1
        if present:
            self.__attendance += 1

    def calculate_attendance(self):
        if self.__total_classes == 0:
            return 0
        return (self.__attendance / self.__total_classes) * 100
```

>_SAMPLE EXECUTION OUTPUT

```
===== Attendance System =====
Student ID: STU-101
Name: Alex Smith
-----
Class 1 Present: yes
Class 2 Present: yes
Class 3 Present: no
-----
Present: 2 | Total: 3
Attendance Rate: 66.67%
```

Explanation: Before vs. After System Impact

✗ Before (Manual System)

Time Consuming: Daily physical roll calls waste class teaching time.

Math Errors: Manual percentage division leads to inaccuracies.

Data Security: Paper records are easily damaged or unauthorizedly modified.

✓ After (Automated OOP)

Instant Logging: Quick prompt entry streamlines attendance logging.

Zero Delay: Python methods calculate exact attendance rates instantly.

Encapsulated Data: Private attributes keep student records secure.

7. SUMMARY

Project Summary & Key Takeaways

Streamlining classroom administration through clean Object-Oriented software design.



100% Encapsulation

Private data security



Zero Math Errors

Automated statistics



EdTech Efficiency

Saved teacher hours

Made by : Made, Richie, Benjamin — Thank You!

Image Sources



https://threeheads.works/wp-content/uploads/2025/03/kaboompics_creative-black-woman-freelancer-working-remotely-in-a-modern-workspace-38513-scaled.jpg

Source: threeheads.works